

Asymmetric Monostable Behavior of an Elastic von Mises Truss Under Base Excitation

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OVERVIEW: This study introduces a constant biasing force to transform a symmetric bistable von Mises truss into an asymmetric monostable system under horizontal base excitation. Breaking structural symmetry allows a shift in the bifurcation topology from pitchfork to saddle-node. The critical force required for such a transition is also explicitly identified.

SUMMARY

Bistable systems are conventionally modeled using an elastic von Mises truss to characterize their snap-through buckling behavior between two stable equilibrium states. When subjected to a horizontal base excitation, the stability of a bistable system is significantly altered; specific combinations of the excitation frequency and amplitude can stabilize the previously unstable equilibrium or destabilize other nominally stable ones. Owing to this dynamic phenomenon, the intentionally stabilized and nominally stable equilibria can respectively serve as OFF and ON states in mechanical switching mechanisms. However, employing a symmetric bistable configuration introduces operational ambiguity during the ON phase owing to the coexistence of two nominally stable states. This study addresses this inherent uncertainty by introducing a constant biasing force that transforms a symmetric bistable system into an asymmetric monostable system. Based on a nondimensional governing equation for the damped elastic truss model, the dynamic behavior of the system was analyzed using both numerical time-domain simulations and bifurcation theory. The results indicate that increasing the biasing force induces symmetry breaking, effectively shifting the bifurcation topology of the system from pitchfork to saddle-node. Furthermore, the critical force required for the transition to an asymmetric monostable regime is explicitly identified. The proposed asymmetric monostable design resolves ON-state ambiguity, offering a highly reliable framework for unambiguous mechanical switching.

INTRODUCTION

The study of nonlinear vibration in modern mechanical systems has evolved from preventing structural instability to utilizing nonlinear dynamic characteristics for functional purposes (1, 2). Within nonlinear systems, multistable systems, particularly bistable systems, have continued to garner interest owing to their unique characteristic of having two distinct stable equilibrium states between which the system can be repeatedly and reliably switched. This ability of a bistable system to occupy and alternate between two well-defined configurations makes it attractive to realize a broad range of engineering functions, including the passive holding of mechanical states, controlled snap-through attraction, and binary signal representation (1, 3).

The practical significance of bistable systems is evident in several mechanical engineering fields. In energy harvesting, snap-through transitions between two stable states amplify the response to environmental vibrations and broaden the effective operating bandwidth, thereby improving the energy conversion efficiency compared to the use of linear resonators (1, 2). In deployable and morphing structures, the two equilibrium configurations correspond to stowed and deployed states, which are each stable without continuous external power, allowing large shape changes to be held passively (4, 5). In mechanical computing, bistable elements serve as physical representations of binary states and form the basis of mechanical switches and logic gates (3, 6). Across these applications, because the reliability of the devices directly depends on the clarity of the distinction between the two configurations of the bistable system, a common requirement is that each stable state should be clearly defined and distinguishable.

The elastic von Mises truss is a representative and widely studied model for bistable systems; it exhibits snap-through buckling and bifurcation behavior governed by loading conditions. The geometric simplicity and nonlinear behavior of this model make it a standard system for investigating state transitions in bistable mechanisms (7, 8).

Research on von Mises trusses has covered a wide range of loading and analytical conditions. Early investigations established the static stability and bifurcation characteristics of trusses under vertical and quasi-static loading, clarifying the conditions under which snap-through and loss of equilibrium occur (8, 9). Subsequent work extended the analysis to the dynamic regime: transient and steady-state responses of axially and laterally loaded structures, the influence of damping, and the dependence of the response on initial conditions (7, 10). More recently, studies have been conducted on how time-varying external excitation can reshape the effective potential landscape of multistable structures, including the use of high-frequency excitation to modify the apparent stiffness and equilibrium positions (4, 11). These studies collectively demonstrate that the qualitative behavior of a von Mises truss is highly sensitive to

the form of the applied loading and that dynamic excitation can produce stability characteristics that differ substantially from those predicted by static analysis alone.

Compared with the extensive literature on vertical and high-frequency loading, the effect of horizontal base excitation on the von Mises truss has received relatively less attention. A recent study by Ghoshal et al. (12) addressed this gap by analyzing the response of a damped von Mises truss subjected to high-frequency horizontal base excitation using the method of multiple scales along with experimental validation. They showed that a horizontal base excitation could alter the stability of the equilibrium points and bias their positions, with the resulting steady-state oscillations centered either on the originally stable equilibrium or on the originally unstable equilibrium, depending on the excitation parameters. This configuration constitutes a symmetric bistable system, in which the two stable equilibria are arranged symmetrically about the central unstable equilibrium.

The symmetric arrangement of the two stable states, which is well suited for demonstrating excitation-induced stability transitions, poses a difficulty when the system is intended to operate as a mechanical switch. If the unstable equilibrium is assigned to the OFF state when it is stabilized and to the nominally stable equilibrium in the ON state, the presence of two symmetric nominally stable equilibria means that the ON state corresponds to two distinct oscillation centers rather than a single, uniquely defined configuration. In mechanical logic gates or precision switches, this additional inverted-on state can introduce logical errors, and resolving them typically requires supplementary physical constraints or signal processing measures (3, 6). A switching mechanism in which each logical state corresponds to exactly one physical configuration should therefore be studied.

This requirement can be satisfied by converting a symmetric bistable system into an asymmetric, monostable system. Introducing a constant external force into the symmetric configuration biases the bistable potential and deepens one potential well relative to the other. Furthermore, beyond sufficient bias, the shallow well is eliminated such that only a single stable equilibrium remains (13, 14). Gravity acting on a suspended mass provides a natural, uniformly distributed constant biasing force and is adopted in this study as a suitable example. Accordingly, this study investigated the steady-state behavior of an asymmetric monostable von Mises truss under horizontal base excitation, focusing on how constant bias governs the number and location of stable and unstable equilibria and how the resulting single-valued configuration can serve as an unambiguous mechanical switch.

RESULTS

Static Equilibrium and the Saddle-Node Bifurcation

Figure 1a shows the bifurcation diagram of static equilibria as a function of G at zero base excitation ($Y_0 = 0$); stable branches are drawn as red solid lines, unstable branches as blue dashed lines. At $G = 0$, the symmetric case of Figure 1a, the two stable equilibria sit symmetrically at $X = 0$ and $X = 2$, about the unstable equilibrium at $X = 1$. As G increases, the two stable branches drift apart toward higher displacement while the unstable branch shifts monotonically toward lower displacement. Beyond a critical value of G , the unstable branch collides with one of the stable branches and both disappear together, which is characteristic of a saddle-node bifurcation. Past this point, one well vanishes entirely and the system changes from bistable to asymmetric monostable.

Symmetric vs. Asymmetric Potential Energy Landscape

Figure 1b shows the evolution of the potential energy landscape corresponding to the bifurcation sequence shown in Figure 1a, where the shape of the potential function is depicted as a function of X for incrementally increasing G values. When G is relatively small ($G \leq 0.3$), the potential energy curves show the classic double-well shape of a bistable system, with two stable equilibria separated by an unstable one. In the symmetric case ($G = 0$), the two wells are equally deep; in the asymmetric case ($0 \leq G \leq 0.3$), one well (on the left in Figure 1b) is noticeably shallower than the other (at larger displacement). This asymmetry was introduced by the gravitational self-weight of the payload, controlled by the dimensionless gravitational parameter G (Equation 5) and changing the payload magnitude therefore changes the degree of well-depth asymmetry. As G increases further ($G > 0.3$), the potential energy ultimately converges to an asymmetric single-well potential beyond the saddle-node bifurcation point.

Dynamic Response of the Asymmetric System

Figure 2a shows that the asymmetric bistable system (below the saddle-node threshold) retains two distinct stable states: a trajectory started at $X_{\text{ind}} = 0$ settles into the shallow-well equilibrium at $X \approx 0.12$, while one started at $X_{\text{ind}} = 2$ settles into the deep-well equilibrium at $X \approx 2.09$. Figure 2b shows the asymmetric monostable case ($G = 0.5$): regardless of initial displacement, both trajectories converge to the same equilibrium at $X \approx 2.19$, confirming that bistability has been lost.

The phase portraits in Figure 2c give a geometric view of the same behavior. Figure 2c shows two clearly separated spiral attractors near $X \approx 0.12$ and 2.09 , with the deep-well attractor showing a visibly larger transient envelope. Figure 2d shows a single attractor at $X \approx 2.19$, into which both trajectories spiral, consistent with the monostable case.

Base-Excitation-Induced Bifurcation in the Symmetric System

Figure 3 shows the oscillation centers of the symmetric system ($G = 0$) as a function of excitation amplitude Y_0 . For Y_0 below about 1.75 , the system retains its double-well character, oscillating symmetrically about the two original equilibria at $X = 0$ and 2 . Beyond $Y_0 \approx 1.75$, the two oscillation centers merge at $X = 1$ and only a single center persists, a pattern consistent with a pitchfork bifurcation of the oscillation centers.

Figure 4 illustrates this with time-domain and phase-plane data for two representative amplitudes (initial conditions $X_{\text{ind}} = 0.5$ and 1.5 , blue and red respectively). At $Y_0 = 1$ (below threshold), the two trajectories settle into separate limit cycles centered at $X \approx 0.18$ and 1.82 (Figure 4a and Figure 4c). At $Y_0 = 1.8$ (above threshold), both trajectories converge onto a single limit cycle centered at $X = 1$ (Figure 4b and Figure 4d), confirming the pitchfork bifurcation identified in Figure 3.

Asymmetric Monostable System Under Base Excitation

Figure 5 shows oscillation centers vs. Y_0 for asymmetric systems with (a) $G = 0.2$ and (b) $G = 0.375$. Nonzero G replaces the pitchfork bifurcation of the symmetric case with a saddle-node bifurcation, and the resulting asymmetry grows with G . At the lower loading ($G = 0.2$, Figure 5a), a saddle-node point still separates a bistable low-amplitude region from a monostable high-amplitude region. At the higher loading ($G = 0.375$, Figure 5b), the bistable region disappears entirely, and the system is monostable across the full amplitude range tested.

Figure 6 shows the corresponding time-domain behavior for $G = 0.2$ and 0.345 . In the bistable case (Figure 6a), a single low-amplitude oscillation state coexists with two distinct high-amplitude oscillation states. In the monostable case (Figure 6b), both the low- Y_0 state ($Y_0 = 1.0$, high equilibrium position) and the high- Y_0 state ($Y_0 = 1.8$, low equilibrium position) converge to a single, uniquely defined oscillation center. One practical caveat: the low-amplitude state in Figure 6b is prone to a high-amplitude inter-well excursion when started from a small initial displacement, which is why the blue-line trajectory was given an initial displacement of 0.7 .

Frequency-Domain Sensitivity

Figure 7 examines oscillation centers as a function of excitation frequency ratio Ω at fixed $Y_0 = 1.5$, for (a) $G = 0.02$ and (b) $G = 0.05$. As with the amplitude sweep, nonzero G replaces the symmetric pitchfork bifurcation with a saddle-node bifurcation. At the lower loading ($G = 0.02$), the bistable region persists over the lower part of the frequency range and transitions to monostable behavior near $\Omega \approx 3.7$. At the higher loading ($G = 0.05$), the bistable region is suppressed across the entire swept frequency range, a value of G roughly seven times smaller than the $G \approx 0.375$ needed to suppress bistability in the amplitude sweep of Figure 7.

DISCUSSION

This study shows that adding a constant biasing force, here the gravitational self-weight of the payload, converts a symmetric bistable von Mises truss into an asymmetric configuration. Consequently, this conversion alters the character of the underlying bifurcation from a pitchfork to a saddle-node type. As the constant biasing force gradually increases, the asymmetric bistable system transitions to an asymmetric monostable state upon crossing the saddle-node threshold. During this process, one of the bistable potential wells progressively diminishes until it vanishes, resulting in a single global equilibrium. Ultimately, this structural change ensures that the nonlinear dynamic trajectories converge toward a sole attractor.

This symmetry-breaking has a specific, practical motivation. Prior work (12) showed that horizontal base excitation alone can drive a symmetric bistable system between a double-well and a single-well dynamic regime, and that the amplitude- or frequency-controlled transition can be used to build a mechanical analogue of an electrical switch: the double-well intra-well state as "ON" and the single-well state as "OFF," with chains of such switches potentially forming mechanical AND/OR logic gates. However, this symmetric implementation has a fundamental drawback: because the two mirror-symmetric potential wells are equivalent, the ON state is not unique but corresponds to two different possible oscillation centers, which introduces ambiguity for any logic-level readout. The gravitationally biased, asymmetric monostable configuration developed in this study resolves this problem: both the ON state and the OFF state converge to a single, globally stable, uniquely defined oscillation center, eliminating the multi-ON-state ambiguity inherent to the symmetric design. The frequency-sweep results add a further practical insight: the gravitational loading needed to eliminate bistability is much smaller when the switching is controlled by frequency rather than by amplitude, suggesting that a frequency-controlled implementation could achieve reliable, unambiguous switching with a substantially weaker bias force, which would relax the design constraints on payload weight or other biasing hardware.

At the same time, several idealizations in the underlying model should be kept in mind when interpreting these results. The governing equations were derived using small-angle approximations, assuming an excitation amplitude small relative to the truss width, frictionless supports, and strictly one-dimensional motion. Effects not captured by the model, including joint friction, material hysteresis, and out-of-plane buckling, could shift the energy threshold required for snap-through behavior in a physical device. Although beyond the scope of this study, the potential for undesirable high-amplitude interwell motion indicates that initial conditions must be strictly bounded in practical switch implementations. In addition, because the asymmetry in this study comes specifically from gravity, the resulting mechanism is inherently orientation-dependent, which could limit its usefulness in contexts such as space structures, where gravitational loading is unavailable or not fixed in direction.

Despite these limitations, the passive, bias-based approach explored here offers a practical design principle for mechanical logic gates and switches that avoid the multi-state ambiguity of symmetric designs without requiring continuous external power or active signal processing. The clean, single-valued separation between ON and OFF attractors should make such switches more reliable in vibration-rich environments, which is relevant to applications such as autonomous morphing structures and vibration-based energy harvesters. Building on these theoretical results, the natural next step is empirical validation on a physical testbed, where real-world nonlinearities, damping variability, and start-up transients can be characterized directly. It would also be worth exploring biasing mechanisms that do not depend on orientation, for example magnetic forces, engineered asymmetric stiffness distributions, or built-in geometric imperfections, as a way to generalize this switching concept beyond gravity-biased designs.

MATHEMATICAL MODELING

Model Description

Figure 8 shows the schematic of the elastic von Mises truss model in stable equilibrium. At this instant, the two springs are at their natural length. The tilted springs are linear, with a stiffness of k and an angle of θ_0 to the horizontal. One end of the spring is connected to the mass, and the other end is fixed to the pivot point on each side. The width between the two pivot points is $2w_0$. The mass is assumed to be focused at one point as a concentrated mass with a value of m . The mass is connected to a viscous damper above with a damping constant of c . Hence, the mass can only move vertically, and the vertical displacement is represented by x as a function of time. The downward direction of the system is considered positive. Gravity under this condition is non-negligible and affects the point mass from its center of mass in the downward direction. The left

pivot point of the system undergoes an excitation, with its displacement represented by y as a function of time with a period of $2\pi/\omega$.

Mathematical Representation

As shown in the free body diagram of Figure 8, when the system moves, the mass undergoes displacement, as shown by the change in the value of $x(t)$, and the left pivot moves, resulting in a change in position shown through $y(t)$. At time t , the deformed lengths of the springs are denoted by Δl_1 (left spring) and Δl_2 (right spring). Using the Pythagorean theorem and subtracting the original length, the deformed spring lengths can be expressed as follows:

$$\begin{aligned}\Delta l_1 &= \sqrt{(w_0 + y)^2 + (w_0 \tan \theta_0 - x)^2} - \sqrt{w_0^2 + (w_0 \tan \theta_0)^2} \\ \Delta l_2 &= \sqrt{w_0^2 + (w_0 \tan \theta_0 - x)^2} - \sqrt{w_0^2 + (w_0 \tan \theta_0)^2}\end{aligned}\quad (\text{Equation 1})$$

The original length of the springs creates sufficient force to balance the force of gravity, making the mass to rest at that instant. By assuming small-angle approximations and applying a Taylor-series expansion to one term, the spring lengths can be approximated in a simpler form.

$$\begin{aligned}\Delta l_1 &= \frac{y^2}{2w_0} + y + \frac{x^2}{2w_0} - \theta_0 x \\ \Delta l_2 &= \frac{x^2}{2w_0} - \theta_0 x\end{aligned}\quad (\text{Equation 2})$$

The instantaneous angles made by the springs can be expressed as follows:

$$\begin{aligned}\theta_1 &= \frac{\theta_0 w_0 - x}{w_0 + y} \approx \frac{\theta_0 w_0 - x}{w_0} \\ \theta_2 &= \frac{\theta_0 w_0 - x}{w_0},\end{aligned}\quad (\text{Equation 3})$$

This is assuming small-angle approximations, and the amplitude of the horizontal excitation of y is considerably lower than the width of the truss.

Thus, the governing equation of the system can be derived:

$$m \frac{d^2 x}{dt^2} = k \Delta l_1 \theta_1 + k \Delta l_2 \theta_2 - c \frac{dx}{dt} + mg \quad (\text{Equation 4})$$

Because the mass is constrained to vertical motion, the horizontal components of the spring forces are balanced by the reactions of the pin support and do not contribute to the equation of motion. The pins are assumed to be frictionless such that each spring rotates freely about its pivot.

Nondimensionalization

To analyze the effects of the parameters on the system with generality, the parameters were nondimensionalized as follows:

$$T = t \cdot \frac{1}{\theta_0} \sqrt{\frac{k}{m}}, X = \frac{x}{\theta_0 w_0}, Y = \frac{y}{\theta_0 w_0}, \Omega = \theta_0^{-1} \cdot \frac{\omega}{\omega_n}, \zeta = \theta_0 \cdot \frac{c}{c_n} = \theta_0 \cdot \frac{c}{2m\omega_n}, G = \frac{mg}{\theta_0^3 w_0 k}, \quad (\text{Equation 5})$$

Here, T , X , Y , Ω , ζ , and G denote the nondimensionalized time, displacement of x , displacement of y , frequency ratio, damping ratio, and constant force (gravity), respectively. The amplitude of the base excitation Y_0 is restricted to $Y_0 < \theta_0^{-1}$. This is because of the previous assumption that the amplitude of the excitation is considerably lower than the truss width.

Hence, the governing equation can be rewritten in a nondimensionalized form:

$$\frac{d^2 X}{dT^2} + 2\zeta \frac{dX}{dT} + X^3 - 3X^2 + \left(\frac{Y_0^2}{2} \cos^2(\Omega T) + \frac{Y_0}{\theta_0} \cos(\Omega T) + 2 \right) X - \frac{Y_0^2}{2} \cos^2(\Omega T) - \frac{Y_0}{\theta_0} \cos(\Omega T) - G = 0 \quad (\text{Equation 6})$$

The equilibrium states of the unforced system when $Y_0 = 0$ and $G = 0$ can be given by $X = 0, 1, 2$. $X = 0, 2$ corresponds to the two nominally stable equilibria, and $X = 1$ corresponds to the unstable equilibrium to be stabilized by the horizontal base excitation.

For the initial conditions of the system, the truss is initially displaced to position X_{ind} and held at rest before being released. The truss is then immediately released when the base is excited, which occurs when $T = 0$. Consequently, the initial conditions are given by

$$X(0^+) = X_{ind} \text{ and } \dot{X}(0^+) = 0 \quad (\text{Equation 7})$$

Unless otherwise stated, $X_{ind} = 2$ is used throughout the paper.

METHODOLOGY

Time Response of a Bistable System

The time response of the system can be easily obtained by numerically integrating the governing equation (Equation 6) with the initial conditions expressed in Equation 7. By introducing the state vector $\mathbf{x} = [x_1 \ x_2] = [\dot{X} \ X]$ into Equation 6, the governing differential equation can be rewritten as a system of first-order differential equations:

$$\dot{\mathbf{x}} = \begin{bmatrix} -2\zeta x_1 - x_2^3 + 3x_2^2 - \left(\frac{Y_0^2}{2} \cos^2(\Omega T) + \frac{Y_0}{\theta_0} \cos(\Omega T) + 2 \right) x_2 + \frac{Y_0^2}{2} \cos^2(\Omega T) + \frac{Y_0}{\theta_0} \cos(\Omega T) + G \\ x_1 \end{bmatrix} \quad (\text{Equation 8})$$

The time response of the elastic von Mises truss model is obtained by numerically integrating Equation 8 using the Runge–Kutta method.

Sweeping Amplitudes and Frequency Ratio

Since the primary interest of this paper lies in changing the equilibrium states of the system, the parameters that were changed, except for the forcing force of gravity (G), were the base excitation amplitude (Y_0) and frequency ratio (Ω). This is because these parameters can be easily varied without stopping the system, unlike system properties such as the damping ratio or spring stiffness.

This section focuses on the effects of changes in the amplitude and frequency ratio of the base excitation on the steady-state responses with different G values. The sweeping method was used to analyze the results. Through simulation, the steady-state response of the system was obtained by eliminating the effect of the initial response with sufficient time such that the oscillations were similar. This was ensured by having an absolute difference in the maximum values of the system lower than 10^{-8} . As the system exhibited a bistable behavior across certain parameter ranges, the process was repeated twice with differing initial conditions biased toward the upper and lower equilibria. This process was repeated for different Y_0 values through forward sweeping with a time response to obtain a continuous set of steady-state responses with increasing Y_0 in increments of 10^{-4} . The same process was repeated with different values of Ω ; however, this was done through backward sweeping with decreasing frequency ratio over time with an increment of 10^{-3} . Finally, sweeping for the steady-state responses was repeated with different G values to determine the effect of the external forcing force on the system. With the opposite sweeping direction, a high-amplitude oscillation of the inter-well motion could be observed; this section only discusses low-amplitude oscillations to evaluate the impact of the gravitational factor.

For amplitude sweeping, $\zeta = 0.05$, $\theta_0 = 30^\circ$, $\Omega = 5$, $X_{ind} = 0.5$ and 1.5 were used under $0.1 \leq Y_0 \leq 3$. Note that the valid range of the amplitude under the mathematical modeling assumptions for such value is $Y_0 < \theta_0 \approx 1.91$. For frequency sweeping, $\zeta = 0.05$, $\theta_0 = 30^\circ$, $Y_0 = 1.5$, $X_{ind} = 0.5$ and 1.5 were used under $2.3 \leq \Omega \leq 5$.

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346 Figures and Figure Captions

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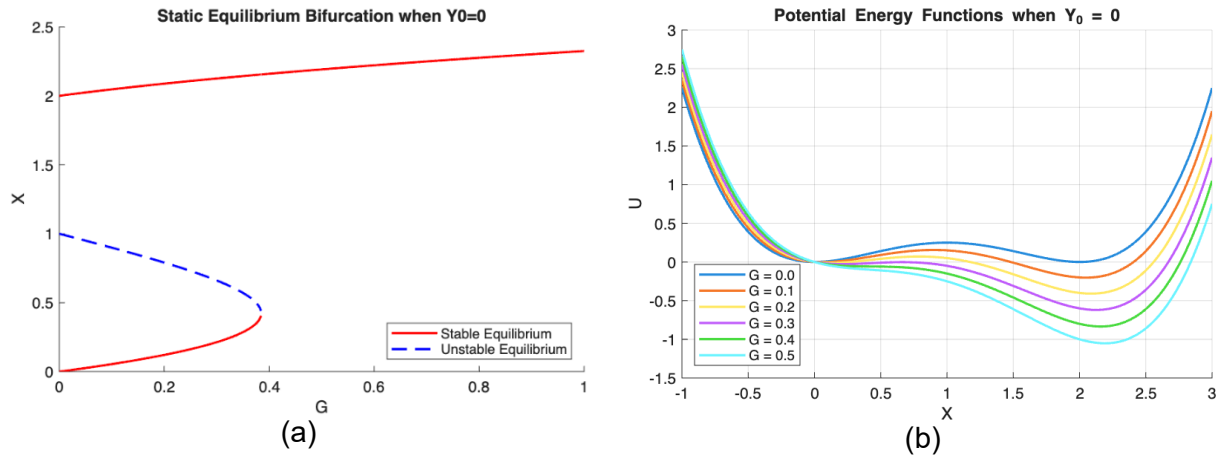


Figure 1. (a) Bifurcation diagram of the static equilibrium solutions as a function of the gravitational parameter G , obtained under zero base excitation. In (b), the evolution of potential energy functions of the bistable oscillator system under zero base excitation with respect to G is demonstrated.

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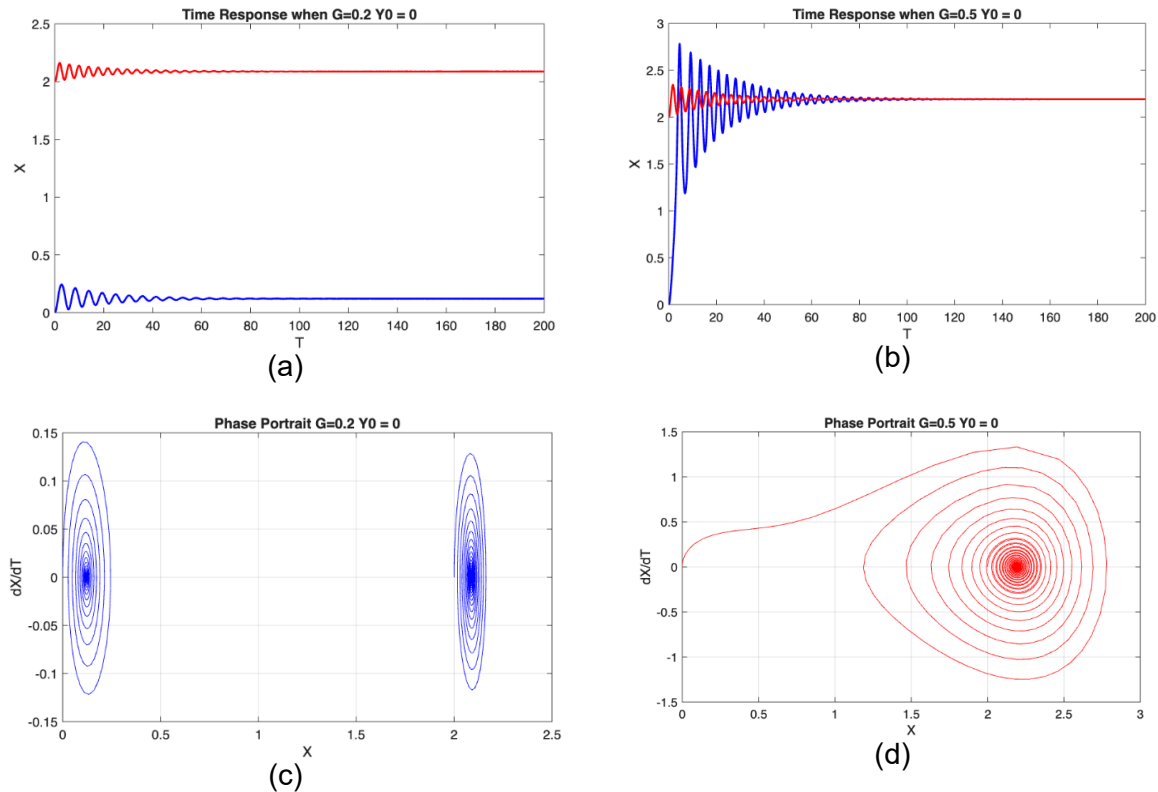


Figure 2. Time responses of the asymmetric (a) bistable and (b) monostable systems, obtained with initial conditions of $X_{ind} = 0$ and 2 (indicated by the blue and red lines, respectively). The phase-plane portraits of the trajectories (a) and (b) are given in (c) and (d), respectively.

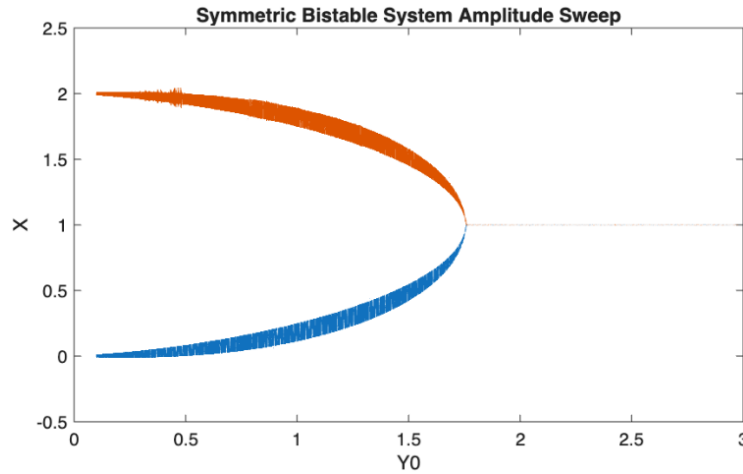


Figure 3. Variation in the oscillation of the symmetric bistable system with respect to excitation amplitude.

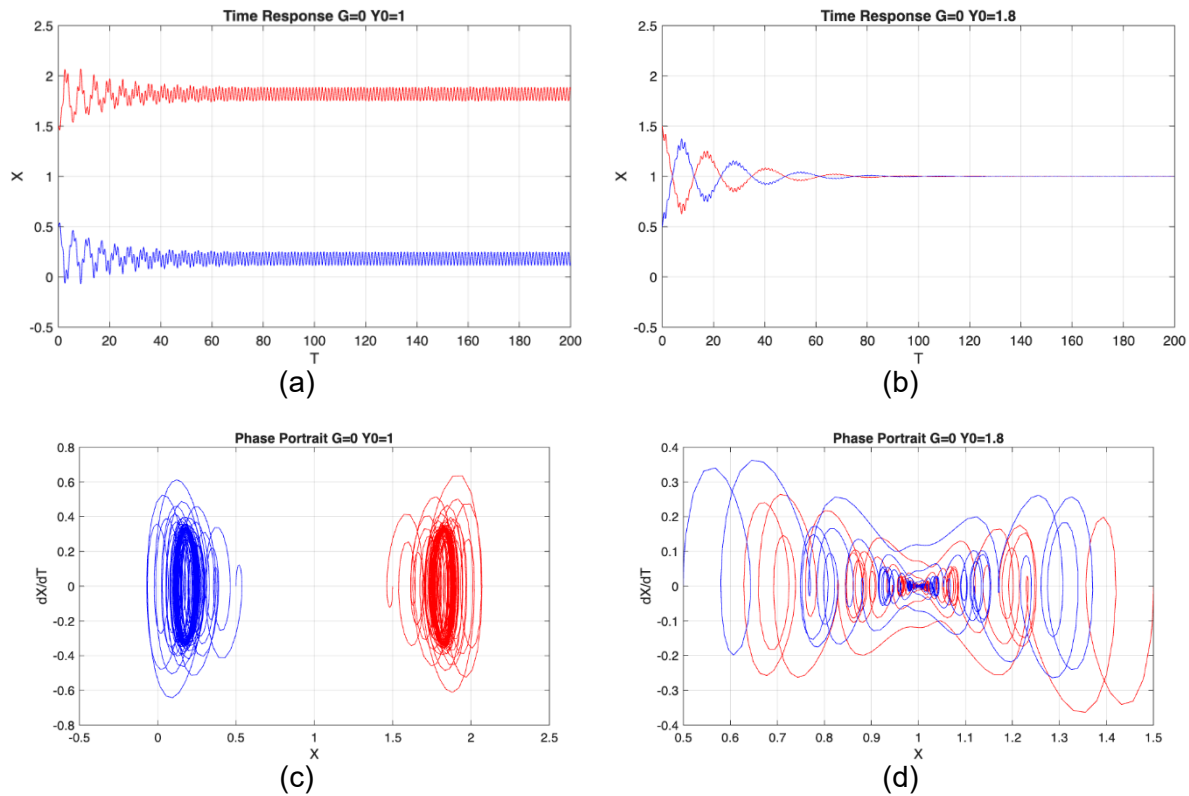


Figure 4. Time-domain responses of the symmetric bistable system ($G = 0$) for two representative excitations: (a) $Y_0 = 1$ and (b) $Y_0 = 1.8$. The phase-plane portraits of the trajectories (a) and (b) are given in (c) and (d), respectively.

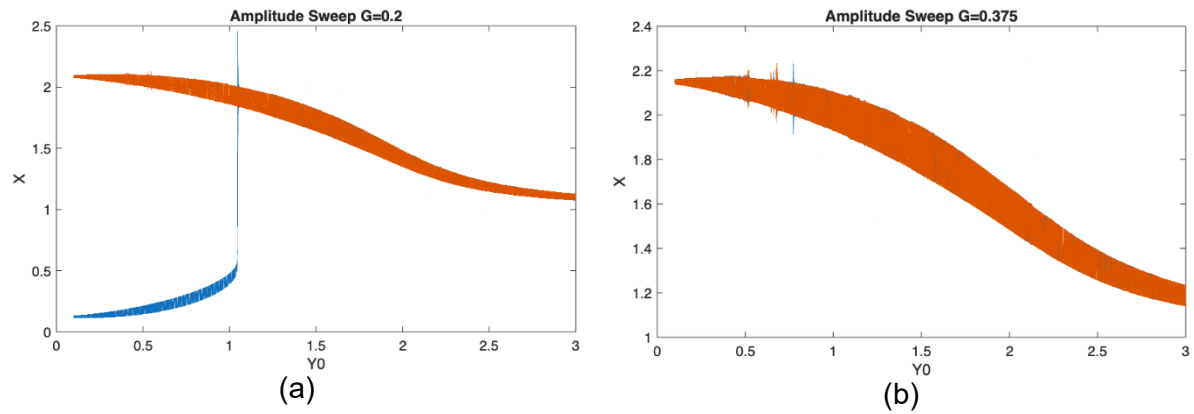


Figure 5. Variations in the oscillations of the asymmetric (a) bistable and (b) monostable systems, obtained for $G = 0.2$ and 0.375 , respectively, with respect to the base-excitation amplitude Y_0 .

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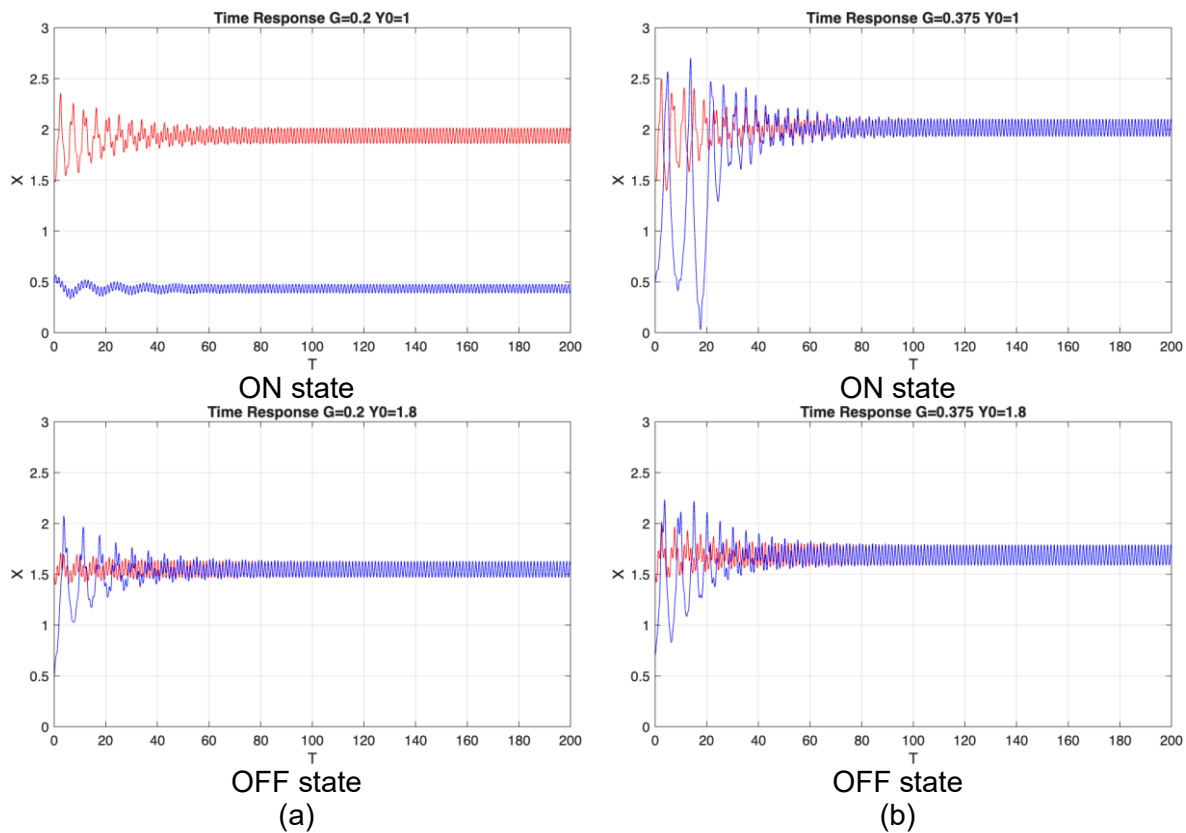


Figure 6. Time-domain responses of the asymmetric (a) bistable and (b) monostable systems for two representative excitations: $Y_0 = 1$ and $Y_0 = 1.8$. The results are obtained with two different initial displacements $X_{ind} = 0.5$ and 1.5 , indicated by the blue and red lines, respectively, except for the OFF state shown in Fig. 10(b), which will be further explained.

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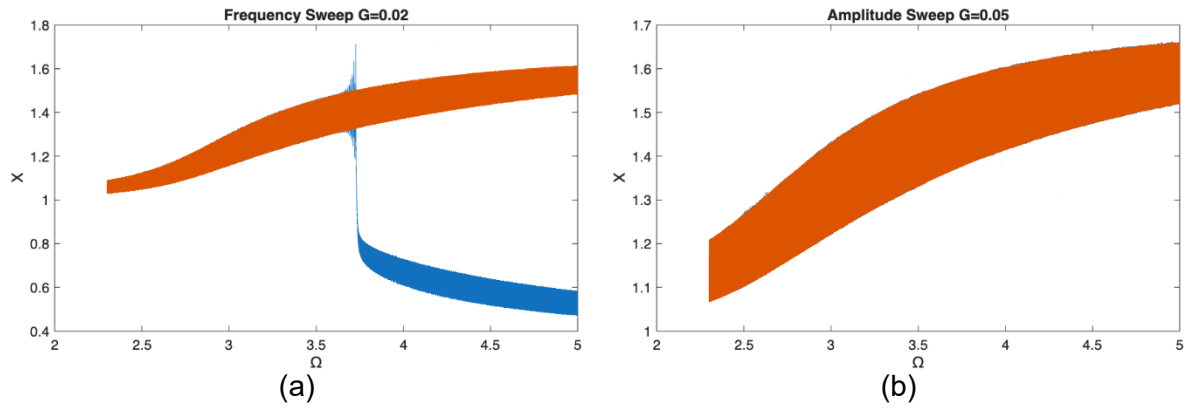


Figure 7. Variations in the oscillations of the asymmetric (a) bistable and (b) monostable systems, obtained for $G = 0.02$ and 0.05 , respectively, with respect to the base-excitation frequency.

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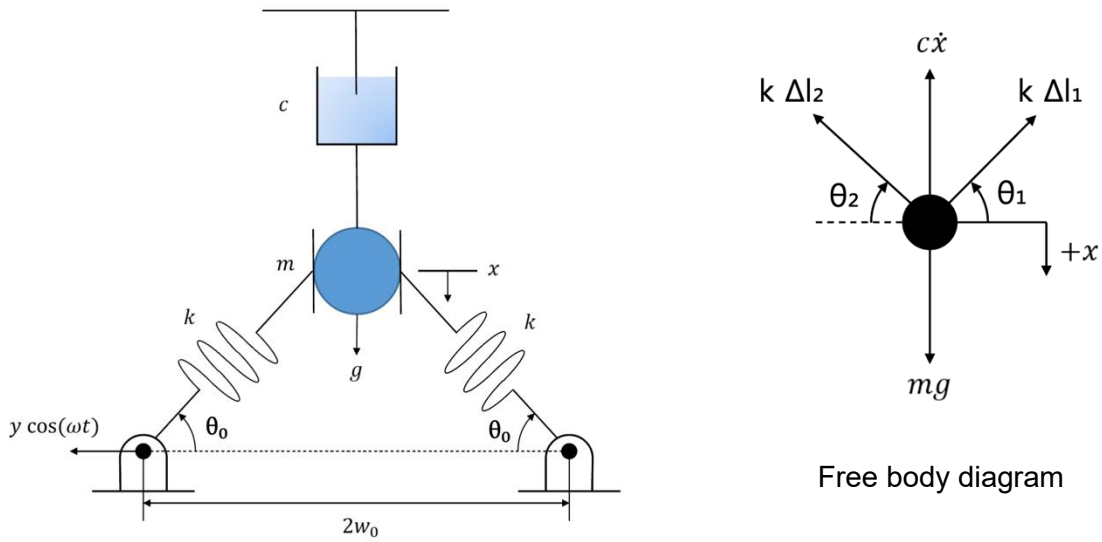


Figure 8. Elastic von Mises truss with viscous dashpot under base excitation and gravitational loading.

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